

 quarter = 4 equal size pieces.

1 litre = 1.06 ^{cuarto} quarts

4 quarts = 1 gallon

rounded to 10th how many litres per 1 gallon.

$0.943 \text{ litre} = 1 \text{ quart}$

$3.77 \text{ litres} = \text{gallon}$

4 quarts = 1 gallon

"a" litres = 1 quart
1 litre = 1.06 quarts

$$a = \frac{1}{1.06} = 0.943$$

how much = 4.24 quarts.
1.06 quarts = 4L
1.06 quarts = 3L
1.06 quarts = 2L
1.06 quarts = 1L

1 gallon

1 litre = 1.06 quarts
a litre = 1 quart

$$\begin{aligned} a \text{ litre} &= 1 \text{ quart} \\ 1 \text{ litre} &= 1.06 \text{ quarts} \\ a \text{ litres} &= \frac{1}{1.06} \\ &= 0.943 \text{ litres} \end{aligned}$$

in 1 quart
4 quarts = 1 gallon

$$\begin{aligned} 4 \text{ litres} &= 4 \times 0.943 \dots \\ &= 3.77 \text{ litres (gallon)} \end{aligned}$$

To make a drink

6-8 fl oz
1 pouch

6-8 fl oz per pouch mix

3 cups of water to 4 pouches of mix

1 cup = 8 fl oz
1 cup = 8 fl oz
1 cup = 8 fl oz
24 fl oz
4 pouches of mix

how much water do I need for 4 pouches?

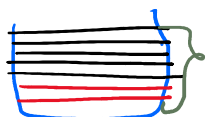
$$1 \text{ cup} = 8 \text{ fl oz}$$

fl = Fluid = liquid
oz = measure = volume.

Unit	Sample Use	Comparing Units
fluid ounce (fl oz)	 1 fl oz	1 fl oz = $\frac{1}{8}$ c
cup (c)	 1 cup 8 fl oz	1 c = 8 fl oz
pint (pt)	 Milk	1 pt = 2 c
quart (qt)	 motor oil	1 qt = 2 pt
gallon (gal)		1 gal = 4 qt

$$\begin{aligned} 24 \text{ fl oz of water} &= 6 \text{ fl oz} \\ 4 \text{ pouches of mix} &= \text{pouch of mix} \end{aligned}$$

$$1 \text{ gallon} = 3.77$$

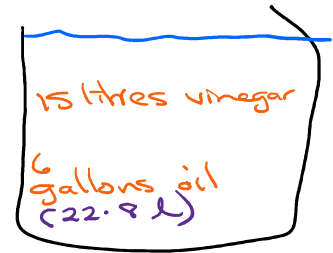


3 x amount

$$\begin{aligned} \text{oil} &: \text{vinegar} \\ 2 \text{ gallons} &= 5 \text{ litres} \\ 6 \text{ gallons} &= 15 \text{ litres} \\ 3 \times 8 \text{ litres} &= 1 \text{ gallon} \\ 22.64 \text{ litres} &= 6 \text{ gallons (oil)} \\ 22.8 & \end{aligned}$$

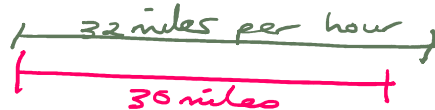
3x amount $\swarrow$ 2 gallons = 5 litres $\swarrow \times 3$ 22.641 litres 6 gallons (oil)
 6 gallons = 15 litres $\swarrow \times 3$ 22.8
 $22.8 \text{ l} = 15 \text{ l}$

Total litres = $15 + 22.8$
 $= 37.8 \text{ litres.}$



Oil Vinegar
 2 gallons = 5 litres
 3x amount $\swarrow \times 3$ 7.54 litre = 5 litres $\swarrow \times 3$
 $22.62 = 15 \text{ litres}$

drive 32 mph average place is 30 miles away
 How long to get there?



miles hour
 32 : 1
 30 : x

hours miles
 $\frac{x}{1} : \frac{30}{32}$

$x = \frac{30}{32}$
 $= \frac{15}{16} \quad 0.9375$
 Round to 1 hour?

$y \text{ mins} = 0.9375 \text{ hours}$
 $60 \text{ mins} = 1 \text{ hour}$

$60 \times \frac{y}{60} = 0.9375 \times 60$
 $y = 0.9375 \times 60$
 $= 56.25 \text{ mins}$

1. \$150 10% off

$150 = 100\%$
 $15 = 10\%$
 $150 - 15 = \$135$

~~150~~ $150 \times 0.9 = 135$

$(150 \times 0.9 = 135)$

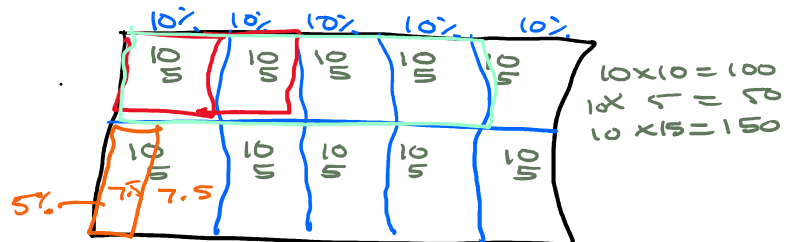
$150 \times 1 = 150$

$90\% = \frac{90}{100}$
 $= \frac{9}{10}$
 $= 0.9$

$100\% - 10\% = 90\%$
 $10\% \text{ of } 60 = 6$
 $60 - 6 = 54$
 $(10\% \text{ off}) = 90\%$
 $10\% \text{ of } 60 = 6$
 $20\% \text{ of } 60 = 12$
 $5\% \text{ of } 60 = 3$
 $10\% \text{ of } 60 = 6$

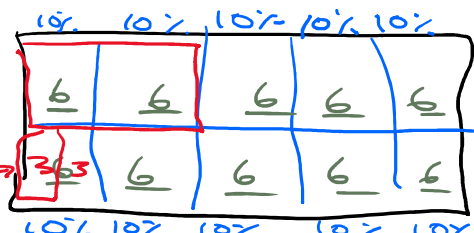
2. What is 10% of \$135

10% of 135 = \$13.50



20% of 150 = 30 5% of 150 = 7.5
 40% of 150 = 60

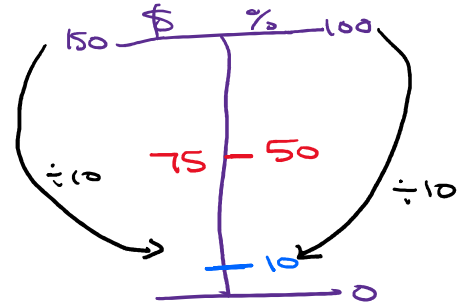
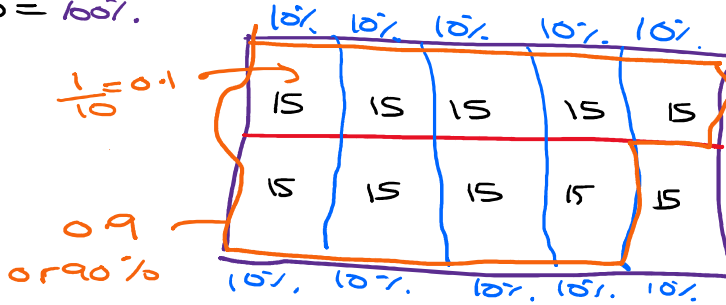
\$60



$$\frac{10}{100} = 0.1$$

$60 - 6 = 54$ 20% of 60 = 12 5% of 60 = 3 10% of 60 = 6
 $(10\% \text{ off}) = 90\%$ 10%, 10%, 10%, 10%, 10%

$$150 = 100\%$$



$$90\% \text{ or } \times 0.9 \Rightarrow 150 \times 0.9 = 135$$

$$\begin{array}{r} 2x - 3 \\ 2 \overline{) 2x - 6} \\ \underline{2x - 6} \\ 0 \end{array}$$

$$2(x - 3) + 4 = 3(x + 2) - 5$$

$$2x - 6 + 4 = 3x + 6 - 5$$

$$2x - 2 = 3x + 1$$

$$-2 = x + 1$$

$$-3 = x$$

$$x = -3$$

Expand (get rid of the brackets)

$$2x - 2 = 3x + 1$$

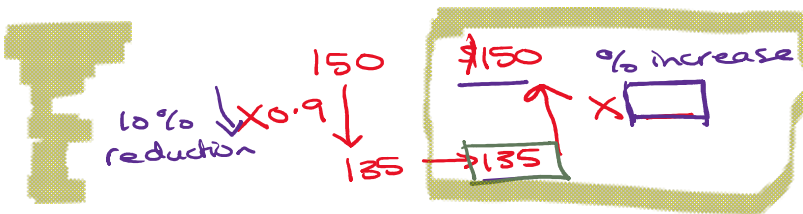
$$2x - 2 = 3x + 1$$

$$-2 = x + 1$$

$$-3 = x$$

$$x = -3$$

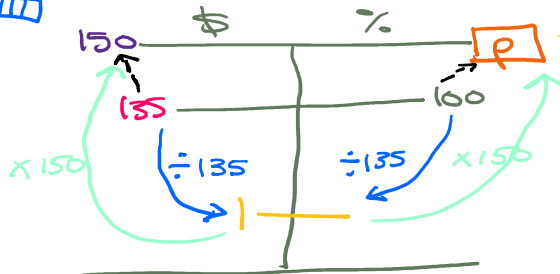
Shop sells \$150, 10% off everything
 new price = \$135 (150 x 0.9 or 150 - 15)
 (90%) 100% - 10%



10% of 135 is 13.5
 $135 + 13.5 = 148.5$
 Not \$150 that I started with!!!
 $\frac{10}{100} = \frac{10}{100} = 90\%$
 $0.9 = 0.90 = 90\%$

$$135 \times \frac{10}{100}$$

10% of 135
 = 13.5
 add 10% of 135
 $135 + 13.5 = 148.5$
 15%
 $135 \times \frac{15}{100} = 20.25$
 So 15% is 20.25
 $135 + 20.25 = 155.25$



$$P = \frac{100}{135} \times 150$$

$$= 111.11$$

it is an 11.11% increase

\$150 → 10% was \$15
 so 10% discount = 150 - 15 = 135

$$90\% \text{ of } 150 = 150 \times 0.9 = 135$$

\$135 what is the % increase to get back to \$150

$$135 \times \frac{0.112}{100} = 150.12$$

111.11% increase

$$135 + 20.25 = 155.25$$

it is an 11.1% increase

$$135 \times \frac{120}{100} = 150$$

$$\frac{150}{135} = 1.111$$

1.111 actually mean

Jacket on Sale! 25% off
original \$18 sale price.

25%	25%	25%	25%
4.50	4.50	4.50	4.50
9	9		
18			

calculate 25%
& subtract

$$18 \times \frac{25}{100} = 4.50$$

$$18 - 4.50 = 13.50$$

multiply to take away 25%

$$18 \times 0.75 = 13.50$$

$$\frac{75}{100} = 0.75$$

$$75\% = \frac{75}{100} = \frac{3}{4}$$

$$\frac{75}{100} = \frac{3}{4} = 0.75$$

25%	25%	25%	25%
20	20	20	20
40	40		
80			

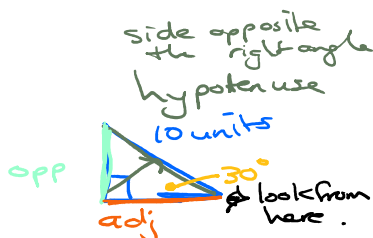
Pants on Sale 25% off

original price is \$80

sale price = ?

25% of 80
discount = \$20
sale price = 80 - 20
= 60

$$80 \times \frac{75}{100} = 80 \times 0.75 = 60$$



opposite = across from
adjacent = next to

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}}$$

$$\tan \theta = \frac{\text{opp}}{\text{adj}}$$

To find the opposite

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$10 \times \sin(30) = \frac{\text{opp}}{10} \times 10$$

$$10 \sin(30) = \text{opp}$$

$$\text{opp} = 5$$

$$\left. \begin{array}{l} \text{opp} = ? \\ \theta = 30 \\ \text{hyp} = 10 \end{array} \right\}$$

To find the adjacent

$$\cos \theta = \frac{\text{adj}}{\text{hyp}}$$

$$\cos(30) = \frac{\text{adj}}{10}$$

$$10 \times \cos(30) = \frac{\text{adj}}{10} \times 10$$

$$\begin{aligned} \text{adj} &= 10 \times \cos(30) \\ &= 5\sqrt{3} \text{ (Exact)} \\ &= 8.66 \text{ (Approx)} \end{aligned}$$

$$\left. \begin{array}{l} \text{adj} = ? \\ \text{hyp} = 10 \\ \theta = 30 \end{array} \right\}$$

You can make only four different cuboids with **16 cubes**.

	Dimensions
Cuboid A 	1 1 16
Cuboid B 	1 2 8
Cuboid C 	1 4 4
Cuboid D 	2 2 4

(a) Which of the cuboids **A** and **D** has the **larger surface area**?
Tick (✓) the correct answer below.

☐ Cuboid A
☐ Cuboid D
☐ Both the same

Explain how you know.

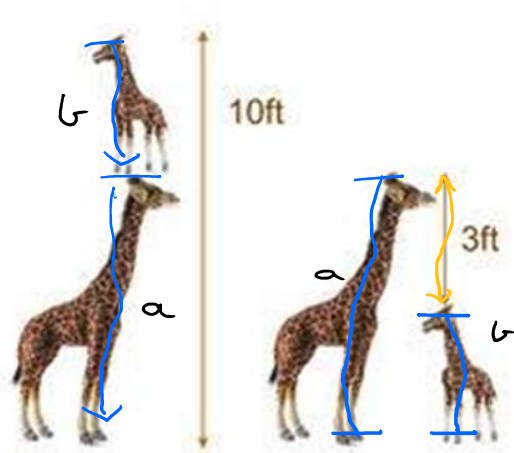
(b) Which cuboid has the **largest volume**?
Tick (✓) the correct answer below.

☐ Cuboid A
☐ Cuboid B
☐ Cuboid C
☐ Cuboid D
☐ All the same

(c) How many of cuboid **D** make a cube of dimensions **4 x 4 x 4**?
.....

(d) You can make only six **different** cuboids with **24 cubes**.
Complete the table to show the dimensions.
Two have been done for you.

	Dimensions
Cuboid E	1 1 24
Cuboid F	1 2 12
Cuboid G	
Cuboid H	
Cuboid I	



find height of each giraffe?
 a = height of the adult giraffe
 b = " " " baby giraffe

$$a + b = 10$$

$$a - b = 3$$

$$a + b = 10$$